

Sentinel

(Artifact Integrity Monitoring)

Project Proposal

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Abstract

Sentinel is a Windows file integrity and tamper-detection tool that tells users when files or build artifacts have changed without authorization. The user registers a trusted folder, and Sentinel records each file's path, size, modification time and SHA-256 hash in a baseline. Later scans compare the folder against that baseline and report modified, deleted, added and moved files. To stop an attacker from hiding changes, the baseline is signed with HMAC-SHA256 and every scan is written to a hash-chained audit log. The project extends this core into CI/CD pipelines: a GitHub Action runs Sentinel on every push and pull request, flags changes to sensitive files such as workflow definitions, Dockerfiles and dependency lockfiles, and fails the build through exit codes. Results are delivered as console output, JSON, an HTML report,a local web-based GUI dashboard and a Discord or Slack webhook alert. Sentinel will be built in Python and completed by 29 November 2026, together with a documented threat model.

1. Introduction

Files on a computer or in a software project can be changed without the owner noticing. Malware may patch a document or program, an insider may edit a configuration file, and an attacker may plant a new script. A cryptographic hash such as SHA-256 gives a simple way to notice this, because even a one-bit change to a file produces a completely different digest [2]. File integrity monitoring builds on that idea: it stores the hashes of a trusted set of files and later compares the current files against them. Tripwire was an early and influential system of this kind [1].

Sentinel brings file integrity monitoring to Windows as a command-line tool with a GUI dashboard. The user registers a folder with INIT, checks it at any time with VERIFY, and approves legitimate changes with ACCEPT. A basic design has an obvious weakness, though: an attacker who can edit the files can also edit the saved baseline or the log of past results. Sentinel closes this gap by signing the baseline with HMAC-SHA256 [3] and recording every scan in a hash-chained audit log, in which each entry contains the hash of the previous one, so removing or altering history can be detected [4].

Modern attacks increasingly target the software supply chain rather than individual machines. Workflow definitions, Dockerfiles, dependency lockfiles and deployment scripts are common targets, because one unnoticed change there can affect every build [5]. For this reason the project extends Sentinel into CI/CD pipelines. A GitHub Action runs Sentinel on every push and pull request, changes to critical files are marked as high severity, and exit codes let any pipeline pass or fail a build automatically.

2. Goals and Objectives

The goal of Sentinel is to detect unauthorized changes to files and build artifacts reliably, and to make that detection trustworthy even when an attacker tries to cover their tracks. The objectives of the project are as follows:

1. To build a command line tool that records a trusted baseline of a folder (path, SHA-256 hash, size and modification time) and reports modified, deleted, added and moved files on every scan.

2. To protect the baseline and the scan history against tampering using an HMAC-SHA256 signature and a hash-chained audit log.
3. To reduce false alarms through ignore rules, move detection and an explicit (ACCEPT) step for authorized changes.
4. To report results in forms suited to both people and machines: a console summary, a JSON export, an HTML report and a Discord or Slack webhook alert.
5. To provide a local web-based GUI dashboard, built with Vue and served by a Python backend, for running scans, reviewing results and browsing scan history without the command line.
6. To integrate Sentinel into CI/CD through a GitHub Action and standard exit codes, so that a pipeline can fail a build when protected files change unexpectedly.
7. To give higher severity to changes in supply-chain-sensitive files such as workflow definitions, Dockerfiles, dependency files, lockfiles and deploy scripts.
8. To document a threat model that states what Sentinel defends against and what it does not.
9. To complete a tested, demonstrable tool by 29 November 2026.

3. Scope of the Project

Sentinel is a Python command-line tool with a local web-based GUI dashboard for Windows that detects unauthorized changes to files and builds artifacts. Its scope covers creating a SHA-256 baseline of a folder (INIT), checking it against that baseline (VERIFY) to report modified, deleted, added and moved files, and approving legitimate changes (ACCEPT), with ignore rules and chunked hashing to keep scans accurate and efficient. To make the results trustworthy, the baseline is signed with HMAC-SHA256 and every scan is recorded in a hash-chained audit log, and the project includes a written threat model. Results are reported through the console, JSON, an HTML report and a Discord or Slack webhook alert, with exit codes that let any pipeline act on them. The project also extends into CI/CD through a GitHub Action that runs on every push and pull request and marks changes to workflow files, Dockerfiles, dependency files, lockfiles and deploy scripts as high severity. A lockfile consistency check, a signed artifact manifest and Windows toast notifications are stretch goals. Real-time monitoring, ACL tracking, GitHub API commit checks,a built-in scheduler and email alerts are out of scope, and the work will be completed by 29 November 2026.

4. Initial Study and Work Done so Far

We explored some existing applications related to the domain. Tripwire established the baseline-and-compare model for file integrity checking [1], and open-source tools such as AIDE follow the same approach on Linux. On Windows, PowerShell's (Get-FileHash) computes a SHA-256 hash for a file, but it does not store a baseline, compare folders, detect moves, protect its own records or integrate with a build pipeline. Sentinel aims to fill that gap with a lightweight Windows tool that also protects its own baseline and logs, and that treats CI/CD files as first-class targets.

We narrowed an initial list of 18 candidate features down to 15 core deliverables and 3 stretch goals, establishing a final schedule that concludes on November 29, 2026. To mitigate risk and protect the timeline, high-risk features; specifically real-time watch mode, ACL tracking and GitHub API commit checks; were explicitly moved out of scope. The execution strategy follows a structured weekly roadmap targeting a strict feature freeze on November 22, 2026. Finally, the project will be presented using a three-part demo script that highlights routine file operations (adding, modifying, and deleting a file), baseline integrity enforcement via HMAC

rejection, and CI/CD policy compliance by demonstrating a failed GitHub Action triggered by a pull request altering a workflow file.

References

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